# Exercise 9.3

Fix $n \geqslant 1$. Consider the naive discretization strategy (9.6), using a finite uniform grid $G = \{y^{(1)}, \ldots, y^{(M)}\}$ over $[-B, B]$. Construct a distribution $P$ and a symmetric score function $s$ such that this approximation has zero marginal coverage for every choice of grid size $M$ and range $B > 0$.

**Solution** First consider the nontrivial range $\alpha \geqslant 1/(n+1)$, with $0 < \alpha < 1$. The very small $\alpha$ case is addressed below. Take $\mathcal{X} = \mathcal{Y} = \mathbb{R}$ and define $P$ by

$$X \sim \mathrm{Unif}(0, 1), \qquad Y = X.$$

Sample $(X_i, Y_i)$ independently from $P$ and choose

$$\boxed{s((x, y); D) = |y - x|.}$$

This score is symmetric because it does not depend on $D$ at all.

For every candidate response $y$, all $n$ training scores are zero:

$$S_i^y = |Y_i - X_i| = 0, \qquad i = 1, \ldots, n,$$

whereas the test score is $S_{n+1}^y = |y - X_{n+1}|$. Since $k = \lceil (n+1)(1-\alpha) \rceil \leqslant n$, the conformal quantile of the training scores is zero. Consequently, the exact full conformal prediction set is

$$\mathcal{C}_{\text{full}}(X_{n+1}) = \{X_{n+1}\},$$

which covers $Y_{n+1} = X_{n+1}$ with probability 1.

Now fix any finite grid $G$. Since $X_{n+1}$ is continuously distributed,

$$P\{X_{n+1} \in G\} \leqslant \sum_{m=1}^{M} P\{X_{n+1} = y^{(m)}\} = 0.$$

Therefore, almost surely every grid point has strictly positive test score and fails the comparison against the zero calibration quantile. The accepted grid-index set is empty. In particular, the expansion of accepted grid points into intervals in (9.6) produces

$$\mathcal{C}_{\text{naive}}(X_{n+1}) = \bigcup_{m \in \varnothing} (y^{(m-1)}, y^{(m+1)}) = \varnothing \quad \text{almost surely.}$$

Hence

$$\boxed{P\{Y_{n+1} \in \mathcal{C}_{\text{naive}}(X_{n+1})\} = 0.}$$

The same $P$ and $s$ work for every fixed finite grid, including every uniform grid of any resolution or range. For $M = 1$, interpret the grid as any chosen single point; the argument is unchanged.

**The necessary restriction on $\alpha$** As written, the exercise omits a restriction needed for its conclusion. If $0 < \alpha < 1/(n+1)$, then

$$\lceil (n+1)(1-\alpha) \rceil = n + 1.$$

The conformal quantile is then $+\infty$, so every grid point is accepted for any real-valued score function. With the boundary convention $y^{(0)} = -\infty$ and $y^{(M+1)} = +\infty$ from Section 9.2.3,

$$\mathcal{C}_{\text{naive}}(X_{n+1}) = \bigcup_{m=1}^{M} (y^{(m-1)}, y^{(m+1)}) = \mathbb{R}.$$

Its coverage is 1, making zero coverage impossible in this range. Thus the counterexample applies precisely when $\alpha \geqslant 1/(n+1)$, or equivalently when the conformal threshold is a finite calibration order statistic. ////